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AI for Bharat Hackathon

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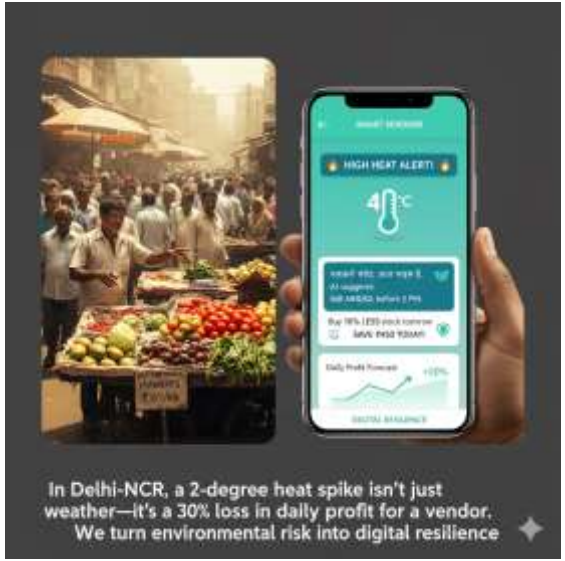
Team Name : Kill-Monster

Team Leader Name : Mahendra Sharma

Problem Statement : SMART VENDORS(REHRI WALA)

Smart Vendors is a voice-first "Decision Intelligence" app. It acts as a virtual business consultant that tells a vendor exactly **what** to buy, **how much** to charge, and **where** to dispose of waste to maximize daily cash-in-hand. It bridges the gap between official government data (Agmarknet) and the chaotic reality of a street corner in local areas of metro city.



THE PROBLEM STATEMENT

The Rotting Loss: ₹500–1,500 lost ost daily to spoliage

THE SOLUTION: with AI Intelligence

The Idea: A voice-first, AI-driven mobile app that predicts, optimizes pricing, pricing, and monetizes waste.

Core Value: It's not s don't know that 5km aneij in worth 20% more.
Information Gap: Vendors dont 5km away in Noida, their stack, their stock is worth 20% more.
Financial Exclusion: No ledger = No bank loans = Predatory 100% interest debt

How is it different?

Unlike **Ninjacart** (which focuses on the supply chain to the retailer) or **SMANN** (which is a B2C delivery app), Smart Mandi focuses on the **Vendor's Internal Profitability**. Most apps require typing and complex navigation; Smart Mandi is **Voice-Only and Multilingual**, designed specifically for low-literacy users in the informal sector.

How will it solve the problem?

The solution uses **Predictive Analytics** to turn environmental risks into business alerts. By combining weather data with market prices, it prevents over-purchasing. To solve the "Fake Data" loophole you mentioned, the app uses a **Consensus Reputation Algorithm**: if a vendor reports an outlier price, the AI cross-references it with nearby peers and official Mandi rates before validating it.

USP (Unique Selling Proposition)

"The Triple-V": Voice, Visuals, and Value.
Voice-First: Zero-typing interface using Bhashini for Hinglish/Bhojpuri.
Visual Freshness AI: A simple photo of the cart gives an "Expiry Countdown."
Waste-to-Wealth: A marketplace that gives "Mandi Credits" for rotted waste, creating a circular economy.

List of features offered by the solution

Hinglish Voice Ledger: Record sales by speaking: *"Bhaiya, 2 kilo tamatar becha ₹80 mein."*

Dynamic Price Pulse: Real-time alerts on localized price changes across Delhi-NCR districts.

Weather-Stock Optimizer: AI-driven buying suggestions based on IMD weather forecasts.

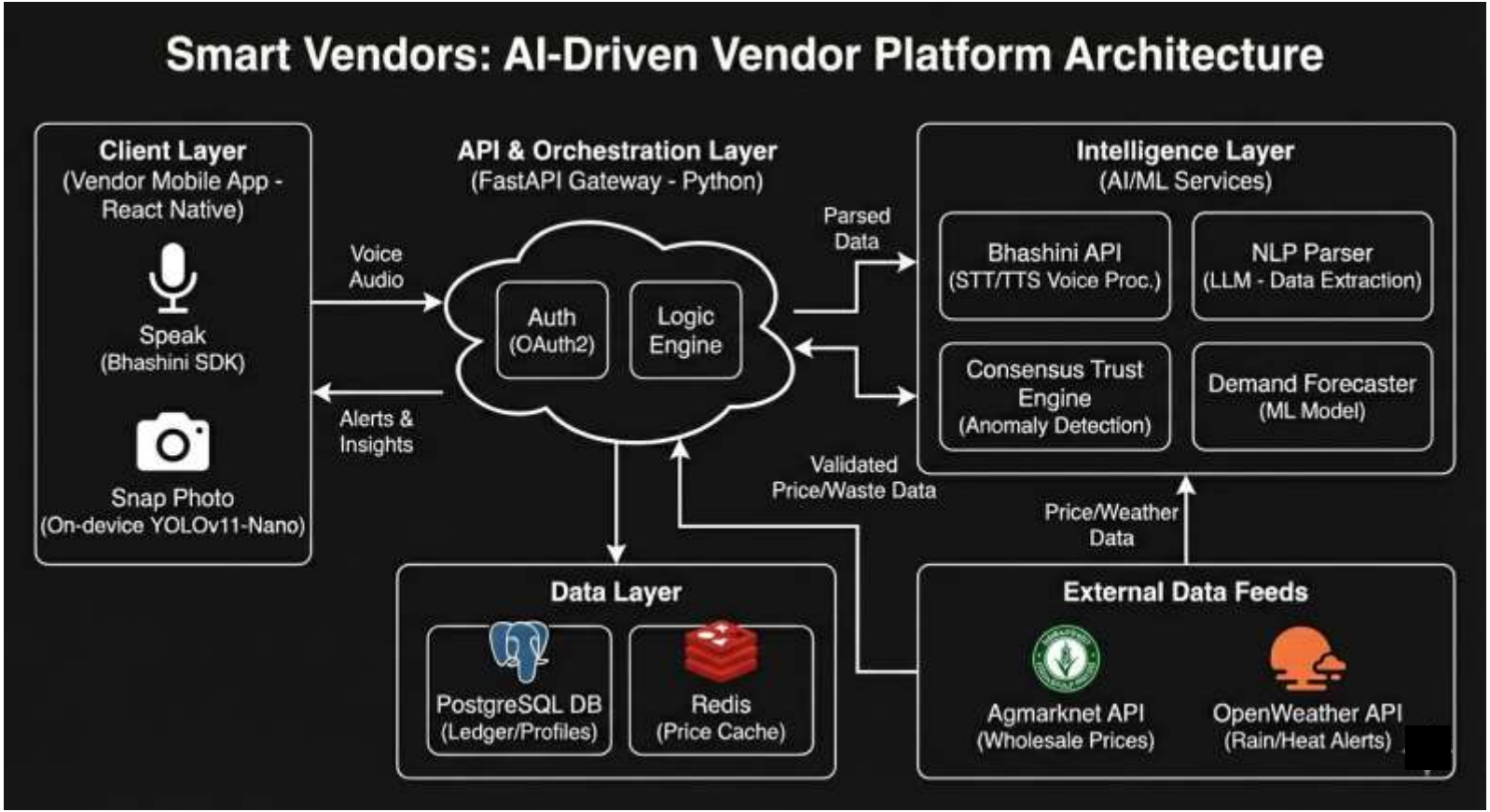
Freshness Scanner: Computer Vision tool to categorize produce as *Fresh, B-Grade (Juice/Pickle), or Waste*.

Waste Value Calculator: Calculates the residual value of rotted goods using the formula:

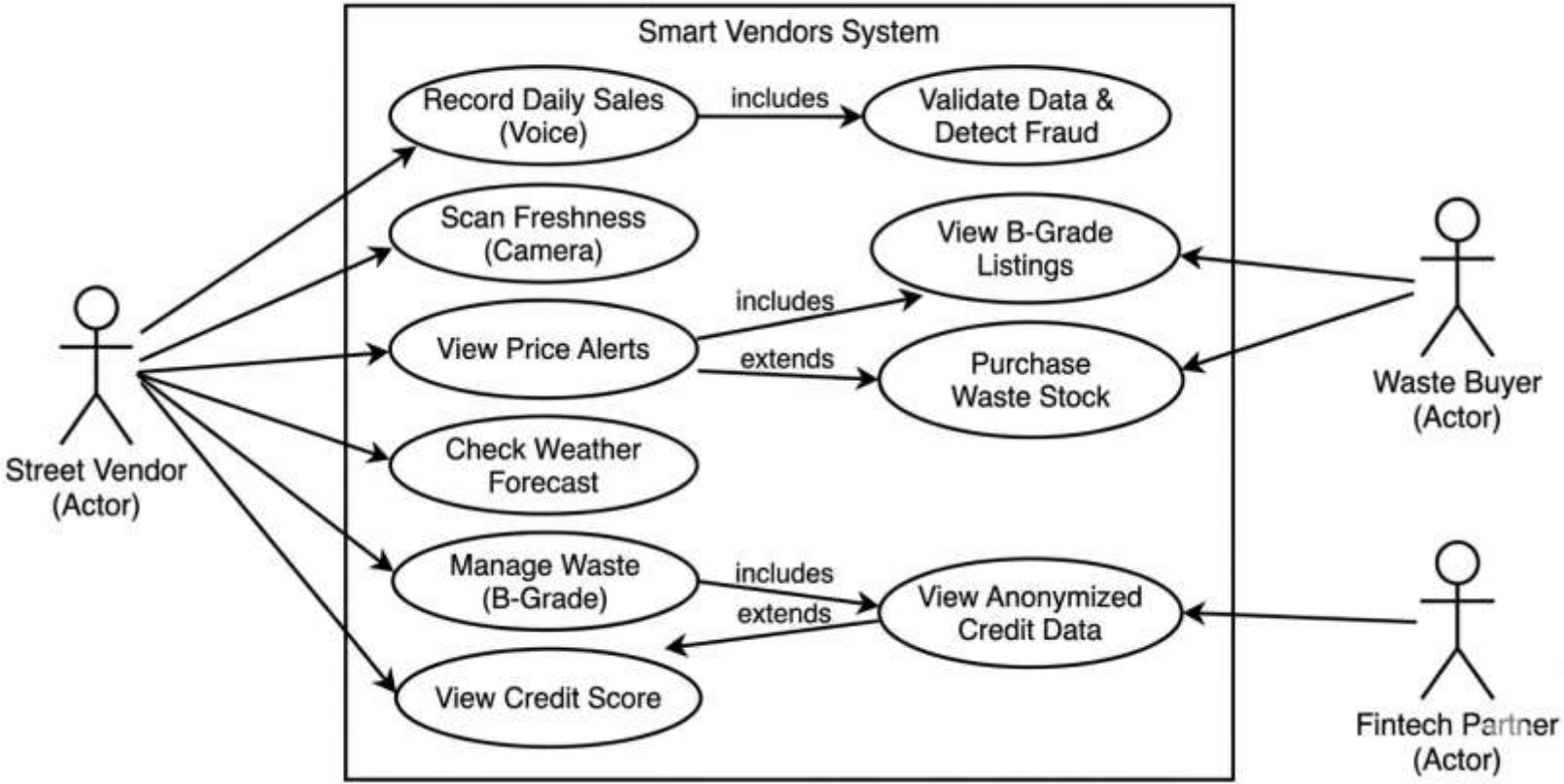
$$V_{residual} = (W \times C_{rate}) - L_{fee}$$

(Where W is weight, C_{rate} is the category rate for compost/biogas, and L_{fee} is the pickup cost).

Micro-Credit Builder: Generates a "Trust Score" based on app usage to unlock small loans options for vendors.


Smart Vendors: AI-Driven Vendor Platform Architecture
Client Layer
(Vendor Mobile App - React Native)
Speak
(Bhashini SDK)
Snap Photo
(On-device YOLOv11-Nano)
Voice Audio
Alerts & Insights
API & Orchestration Layer
(FastAPI Gateway - Python)
Auth
(OAuth2)
Logic Engine
Parsed Data
Intelligence Layer
(AI/ML Services)
Bhashini API
(STT/TTS Voice Proc.)
NLP Parser
(LLM - Data Extraction)
Consensus Trust Engine
(Anomaly Detection)
Demand Forecaster
(ML Model)
Validated Price/Waste Data
Price/Weather Data
Data Layer
PostgreSQL DB
(Ledger/Profiles)
Redis
(Price Cache)
External Data Feeds
Agmarknet API
(Wholesale Prices)
OpenWeather API
(Rain/Heat Alerts)

A Use-Case Diagram for "Smart Vendors"



Technologies to be used in the solution:

Voice: Bhashini API (Natural language for Hindi/Slang).
Vision: YOLOv11 (Lightweight computer vision for produce grading).
Backend: Python (FastAPI) + Google Cloud (for high-availability).
Front-end: React Native

Estimated implementation cost (optional):

Add as per the requirements for the hackathon:

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Thank You